

ANGELA KANG

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EDUCATION

University of Southern California

M.S. Applied Data Science | GPA: 4.0

Los Angeles, CA

August 2025 – May 2027(expected)

• **Relevant Courses:** Machine Learning for Data Science, Foundations of Data Management

University of California, Santa Barbara (UCSB)

Goleta, CA

B.S. Statistics and Data Science

September 2021 – June 2025

• **Relevant Courses:** Probability & Statistics, Machine Learning, Multivariate Analysis, Time Series, Stochastic Processes, Linear Algebra, Real Analysis, Numerical Methods, Data Structures & Algorithms, Experimental Design, Python Programming

EXPERIENCE & LEADERSHIP

Hackathon Winner, SBHacks XI | JavaScript, HTML/CSS, Chrome Extension APIs

• Led front-end development of My Meowcho, a Chrome extension that gamifies productivity with animated cats, built with a 4-person team; won Best Overall Hack out of 60+ teams.

PROJECTS

Scout: AI Freelancer–Client Platform | React, React Native, FastAPI, WebSocket, ChromaDB, GPT-4o-mini

- Freelancers lose money to scope creep when clients slip extra requests into chat. As Product Lead on a 3-person team, built a web and Android platform that catches these requests in real time.
- Built the React (Vite) web app and React Native Android app: real-time WebSocket chat, project dashboard, and conversational onboarding. Designed the UX and high-fidelity mockups for all major screens.
- Built the Scope Guardian alert and change-order UI on top of the team's RAG pipeline, which pulls the 3 most relevant contract clauses for each message and asks GPT-4o-mini whether the request is out of scope.

AI Group Chat | FastAPI, MySQL, SQLAlchemy, JWT, WebSocket, Llama 3

- Built a real-time group chat where an LLM assistant joins the conversation and answers questions for everyone in the room, with JWT login and messages stored in MySQL.
- Connected the bot to any OpenAI-compatible endpoint, so it can run on a local model (llama.cpp or vLLM serving Llama 3 8B) or a hosted API by changing config only; packaged it as an Android app (Trusted Web Activity).

PDF Chat: Document Q&A with RAG | LangChain, FAISS, sentence-transformers, llama.cpp, OpenAI

- Wrote the core app for a 3-person team project: a chatbot that answers questions about long technical manuals by extracting PDF text, splitting it into overlapping chunks, and indexing embeddings in FAISS.
- Supported two swappable backends, OpenAI or fully offline (sentence-transformers + 4-bit Qwen 2.5 3B via llama.cpp), with chat memory so follow-up questions keep their context.

CUDA Matrix Multiplication & Convolution | C, CUDA, cuBLAS, Python ctypes

- Tested how GPU memory strategy affects speed by writing matrix multiply 4 ways (CPU, naive CUDA, shared-memory tiled CUDA, cuBLAS) plus 2D image convolution on an RTX 3060.
- Tiled kernel ran a 2048×2048 multiply in 19.5 ms vs. 58 s on single-threaded CPU (cuBLAS: 2.3 ms); compiled kernels into a shared library callable from Python with NumPy arrays.

Jute Pest Classification | Python, TensorFlow/Keras, Transfer Learning

- Automated identification of 17 jute pest species from ~6,800 images, a task that normally needs field experts.
- Benchmarked 5 ImageNet-pretrained CNNs with the same classification head, data augmentation, early stopping, and mixed-precision GPU training; DenseNet201 reached 96.6% test accuracy (F1 0.97) vs. 83.1% for the next best.

COVID Custom SQL Engine | Python (no pandas), Streamlit, Plotly

- Built core database operations from scratch (CSV parser, filter, projection, group-by, hash join) on a custom column-based table to learn how SQL engines work.
- Ran it on a 350K-row (92 MB) COVID dataset joined with country data and deployed a live Streamlit dashboard of cases, deaths, and vaccination trends.

SKILLS & INTERESTS

Languages: Python, R, SQL, C, CUDA, JavaScript, HTML, MATLAB, LaTeX, C++, React

Data Science & Analytics: Predictive modeling, statistical modeling, regression analysis, time series analysis, experimental design, ANOVA, machine learning, deep learning, transfer learning, exploratory data analysis (EDA), feature engineering, dimensionality reduction (PCA), imbalanced-data methods (SMOTE), clustering, relational data modeling, custom SQL-style operations (filter, select, groupby, join)

Tools & Frameworks: Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, Git/GitHub, Jupyter, Excel, Streamlit, MySQL, MongoDB, Apache Spark, Hadoop, cuBLAS, BeautifulSoup, Selenium, Tesseract OCR, GCC, NVCC

Technical Skills: GPU programming & parallel computing, CUDA kernel development, high-performance computing, Linux/Ubuntu development, web scraping, data pipeline development, database querying, SQL joins and aggregations, NoSQL-style data handling, distributed data processing concepts, Python bindings to C/CUDA code, statistical analysis, data visualization